AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM311
Information Security

Final Examination
Spring Semester 2025/2026

Date: 10/June/2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 3 Hours

Instructions:

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **NOT** allowed.

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PART 1: Multiple Choice Questions [20 Marks]

Questions 1 (2 marks/each):

1. Malware is defined as:
A. Software that improves performance
B. Software inserted to compromise confidentiality, integrity, or availability
C. Only viruses
D. Only spyware

2. A botnet refers to:
A. Secure network of servers
B. Collection of infected computers controlled remotely
C. Firewall system
D. Authentication system

3. Ransomware is characterized by:
A. Stealing CPU power only
B. Locking data and demanding payment
C. Monitoring network traffic
D. Sending spam

4. Websites collect personal data using:
A. Only registration forms
B. Only surveys
C. Cookies and tracking technologies
D. Firewalls

5. Threat modeling is mainly used to:
A. Write code
B. Identify important assets and protect them
C. Test performance
D. Manage databases

6. The main objective of risk assessment is to:
A. Eliminate all risks
B. Increase system speed
C. Determine budget and optimize protection
D. Remove users

7. Loss of autonomy refers to:
A. Behavioral restrictions
B. Financial loss
C. System failure
D. Encryption failure

8. Privacy awareness is:
A. Technical encryption method
B. Understanding privacy importance and responsibilities
C. Firewall rule
D. Malware detection

9. Negligent behavior involves:
A. Intent to harm
B. No decision involved
C. Conscious inappropriate action without harm intent
D. Automated attack

10. A security incident is:
A. Any normal activity
B. Event that compromises confidentiality, integrity, or availability
C. Logging activity
D. User login

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PART 2: Short Answer Questions [30 Marks]

P2-Q1 [6 marks]: Define malware and explain how it spreads.
P2-Q2 [6 marks]: Explain the purpose of a Privacy Impact Assessment (PIA)
P2-Q3 [6 marks]: What are the objectives of security event logging?
P2-Q4 [6 marks]: An organization identifies High impact data breach with Moderate likelihood of occurrence. Explain how risk is determined.
P2-Q5 [6 marks]: An employee shares sensitive data via personal email unintentionally. Classify behavior and Explain why it occurred.

PART 3: Problem Solving Questions [50 Marks]

P3-Q1 [10 marks]: Case Study: An organization notices that several computers are slow and communicating with unknown external servers. Investigation shows that attackers are remotely controlling these machines.
P3-Q1-1 [3 marks]: Identify the type of malware involved.
P3-Q1-2 [4 marks]: Explain how it works.
P3-Q1-3 [3 marks]: Suggest two mitigations.

P3-Q2 [8 marks]: Case Study: A website collects user data via cookies and shares it with third-party advertisers without user knowledge.
P3-Q2-1 [2 marks]: Identify the privacy issue.
P3-Q2-2 [4 marks]: Explain the role of data brokers.
P3-Q2-3 [2 marks]: Suggest one privacy protection measure.

P3-Q3 [14 marks]: Scenario: Smart Hospital Cyberattack (Full Integration)
A smart hospital integrates:

- IoT medical devices (heart monitors, insulin pumps)
- Cloud-based Electronic Health Records (EHR)
- Mobile apps for patients
- Third-party insurance and pharmacy systems

Recently:

- IoT devices sent incorrect readings
- Patient data was encrypted and ransom demanded
- Attackers accessed admin accounts
- Logs show unusual traffic before the attack

P3-Q3-1 [3 marks]: Identify THREE types of malware involved
P3-Q3-2 [2 marks]: Classify TWO privacy harms caused
P3-Q3-3 [4 marks]: Explain how threat modeling could have prevented this
P3-Q3-4 [2 marks]: Identify TWO vulnerabilities exploited
P3-Q3-5 [3 marks]: Propose THREE mitigation strategies

P3-Q4 [8 marks]: Scenario: A smart city system collects Location data, Health data, and Surveillance data. Data is leaked and used for profiling citizens.
P3-Q4-1 [3 marks]: Identify THREE privacy harms
P3-Q4-2 [2 marks]: Explain “loss of self-determination”
P3-Q4-3 [3 marks]: Suggest mitigation measures

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P3-Q5 [10 marks]: Scenario: A government deploys a Smart Emergency Response System (SERS) that integrates:

- Ambulance IoT systems (real-time patient vitals)
- Hospital Electronic Health Records (EHR)
- Police command center
- Citizen emergency mobile application
- Cloud analytics platform
- Third-party insurance system

The System Behavior is described as follows:

- **Ambulance systems** collect and transmit real-time patient data (e.g., vitals, incident notes) and are authorized to create and update emergency requests and initial patient records before hospital arrival.
- **Doctors** can view and update patient medical records after the patient is admitted, including diagnosis, treatment plans, and prescription updates. Doctors are not allowed to delete records.
- **Police officers** can view emergency requests and limited patient-related information strictly for incident coordination purposes. They are not allowed to modify or create medical data.
- **Citizen mobile application** allows users to submit emergency requests and provide initial personal or medical information (e.g., symptoms, location). Citizens may enter or update their own submitted data only, but cannot access or modify full patient records.
- **Cloud analytics platform** processes incoming data to prioritize emergency cases but does not directly modify patient records.
- **Insurance systems** access relevant patient and treatment data through controlled interfaces (not directly exposed in this model).
- **System administrators** have full access to all system components, including logs, records, and configurations, for maintenance, auditing, and system management purposes.
- **System logs** are generated automatically and are accessible only to administrators for monitoring, auditing, and incident investigation.

P3-Q5-1 [5 marks]: Construct an **Access Control Matrix** using the following:

Subjects:

- Doctor
- Ambulance System
- Police Officer
- Citizen Application
- System Administrator

Objects:

- Patient Records
- Emergency Requests
- IoT Device Data
- System Logs

Access Rights:

- Read (R), Write (W), Update (U), Delete (D)

P3-Q5-2 [5 marks]: Convert your matrix into an **Access Control List (ACL)** for: **Patient Records**

End of questions